THE HONG KONG POLYTECHNIC UNIVERSITY
香港理工大學

Group Meeting

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April 29, 2026

Opening Minds • Shaping the Future
啟迪思維 • 成就未來

Framework

Typhoon-induced Risk Assessment of Electricity transmission tower-line System in Gulf topography

- **<u>Typhoon-induced reliability Model</u>**
- The Influence of Gulf topography
- Wind tunnel test of ETTL
- ML response model for Single ETTL
- Fragility Curve of ETTL
- Risk Assessment of ETTL System

Typhoon-induced reliability Model

Wind Field Simulation

MATLAB Code

```
%参考点设定
m=10;                       %模拟风速个数
coord = [
     1,  1,   10;
     1,  2,   28;
     2,  3,   32;
     3,  4,   38;
     4,  5,   44;
     5,  6,   50;
     6,  7,   60;
     7,  8,   65;
     8,  9,   70;
     9,  10,   75];          % 模拟点坐标位置 Unit:m x,y,z

z_length = 10;              %参考点的高度
U_0=50;                     %参考点处的平均风速
```

Normal wind profile
$$U(z)=\frac{u_*}{\kappa}\log\left(\frac{z}{z_0}\right)$$

Kaimal's spectrum[1]
$$\frac{nS_u(\omega)}{u_*^2}=\frac{200f}{(1+50f)^{5/3}}$$

Panofsky's spectrum[2]
$$\frac{nS_w(\omega)}{u_*^2}=\frac{3.36f}{(1+10f)^{5/3}}$$

$$f=nz/U(z)$$

Coherence function
$$S_c(r,n)=\sqrt{S(x_1,n)\cdot S(x_2,n)}\cdot \mathrm{Coh}(r,n)$$
$$\mathrm{Coh}(r,n)=e^{-f}$$
$$f=\frac{n\left[C_z^2(z_1-z_2)^2+C_y^2(y_1-y_2)^2+C_x^2(x_1-x_2)^2\right]^{1/2}}{\frac{1}{2}\left[U(z_1)+U(z_2)\right]}$$

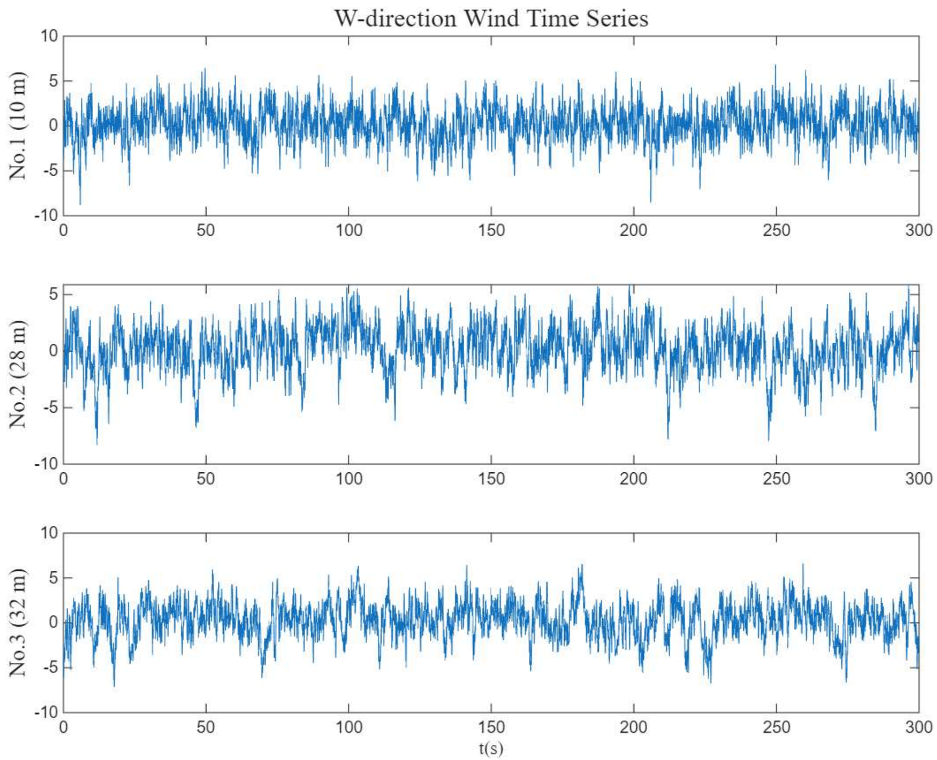
Harmonic synthesis
$$f_j(t)=2\sqrt{n}\cdot\sum_{k=1}^{j}\sum_{l=1}^{M}\left|H_{jk}(\omega_{kl})\right|\cos\left(\omega_{kl}t-\theta_{jk}(\omega_{kl})+\varphi_{kl}\right)$$
$$(j=1,2,3,\cdots,m)$$

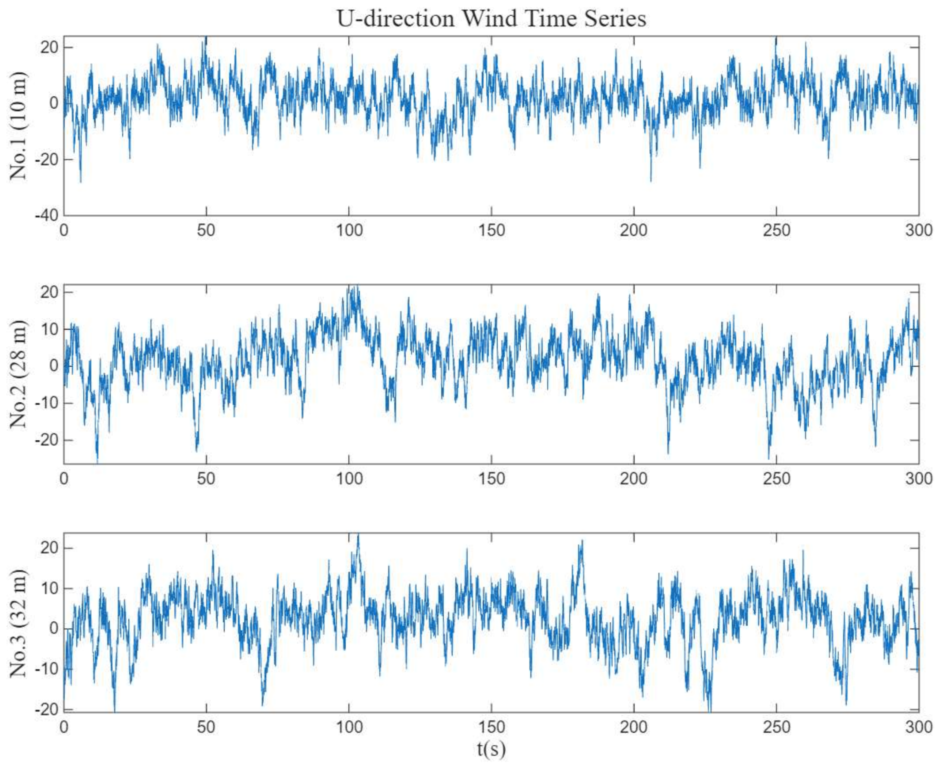
[1] KAIMAL J C, WYNGAARD J C, IZUMI Y, et al. Spectral characteristics of surface-layer turbulence [J]. 1972, 98(417): 563-89.

[2] LUMLEY J L, PANOFSKY H A. The structure of atmospheric turbulence. [J]. New York (Interscience Publishers), 1965, 91(389): 400-.

Typhoon-induced reliability Model

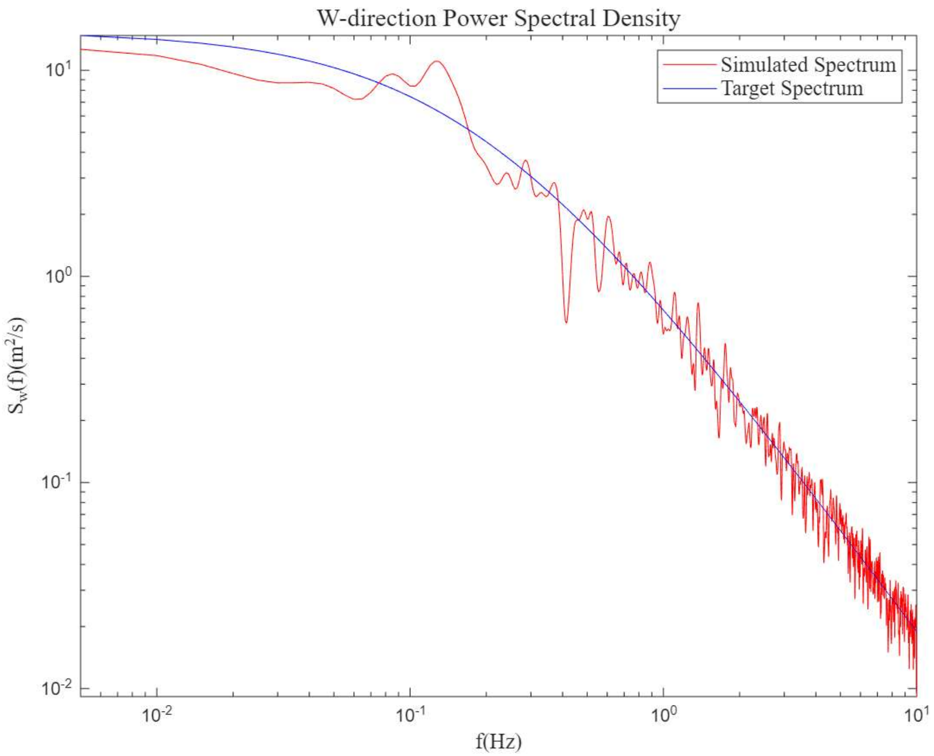
Typhoon-induced reliability Model - Wind Field Simulation

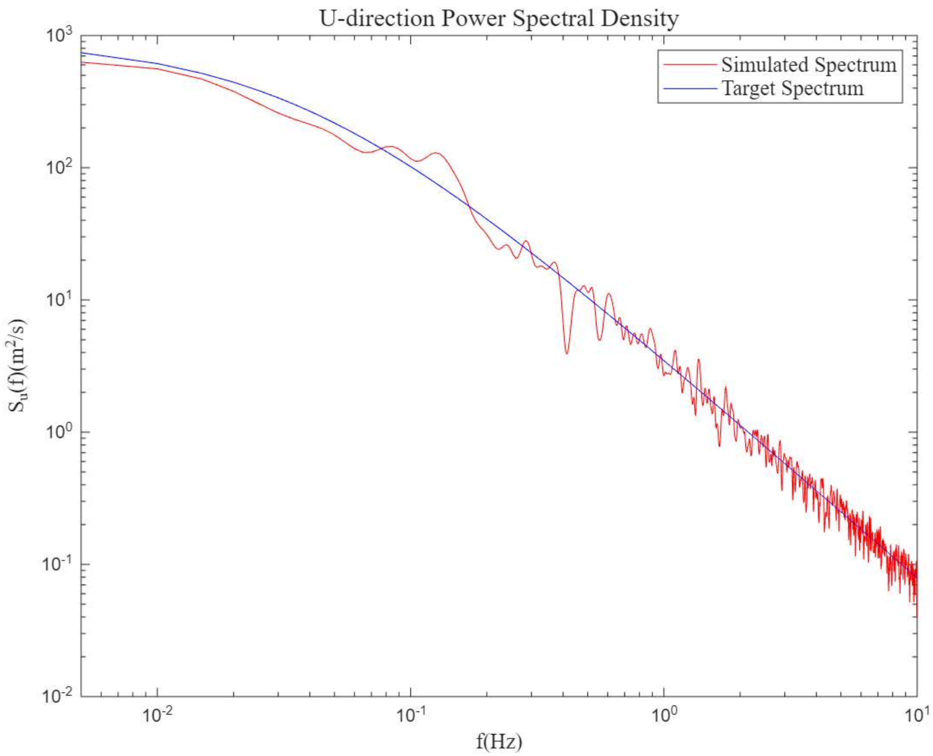




Typhoon-induced reliability Model

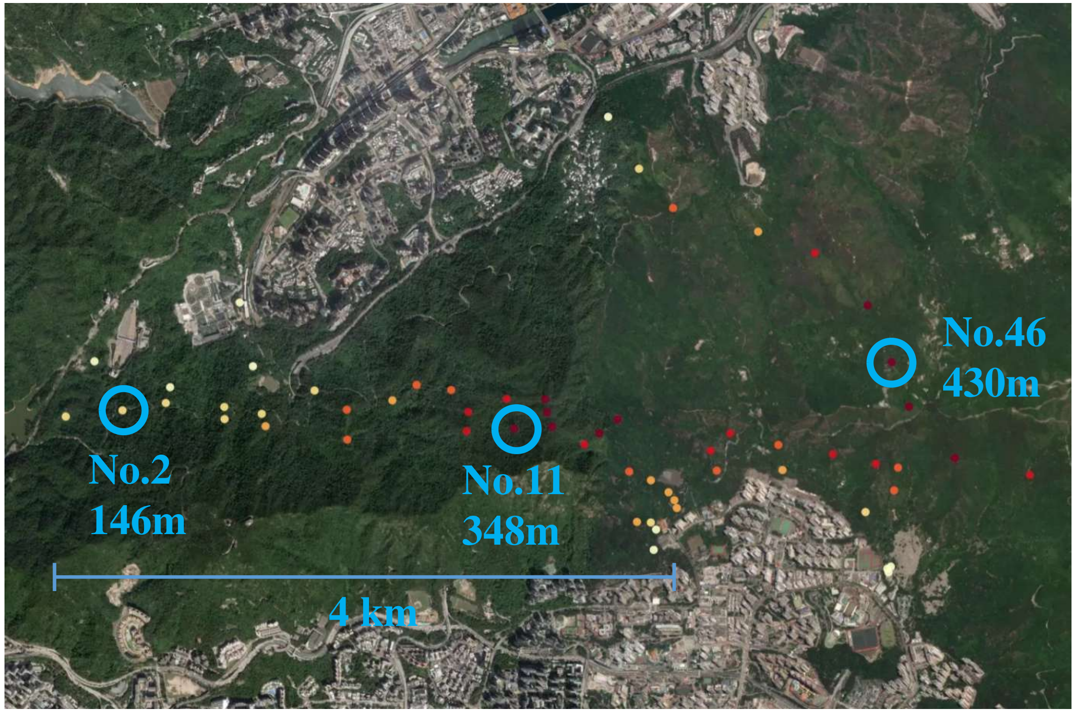
Typhoon-induced reliability Model - Wind Field Simulation

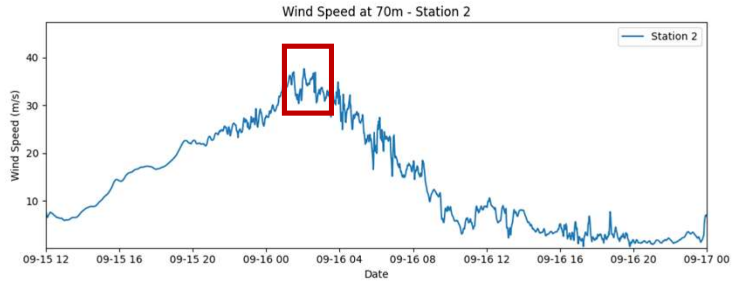


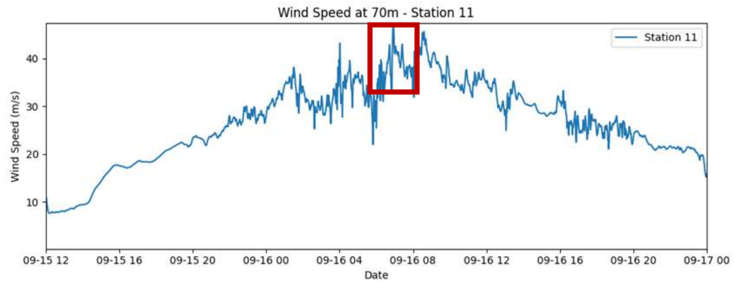


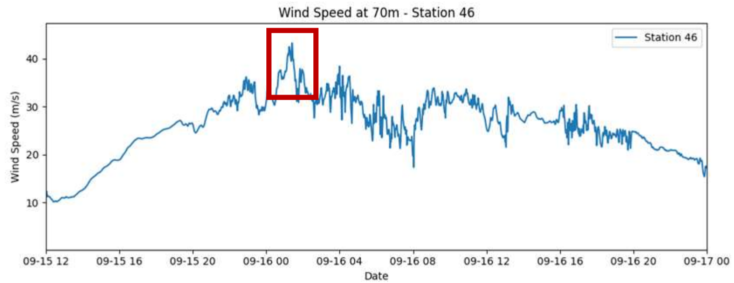
Typhoon-induced reliability Model

Typhoon-induced reliability Model – WRF Model



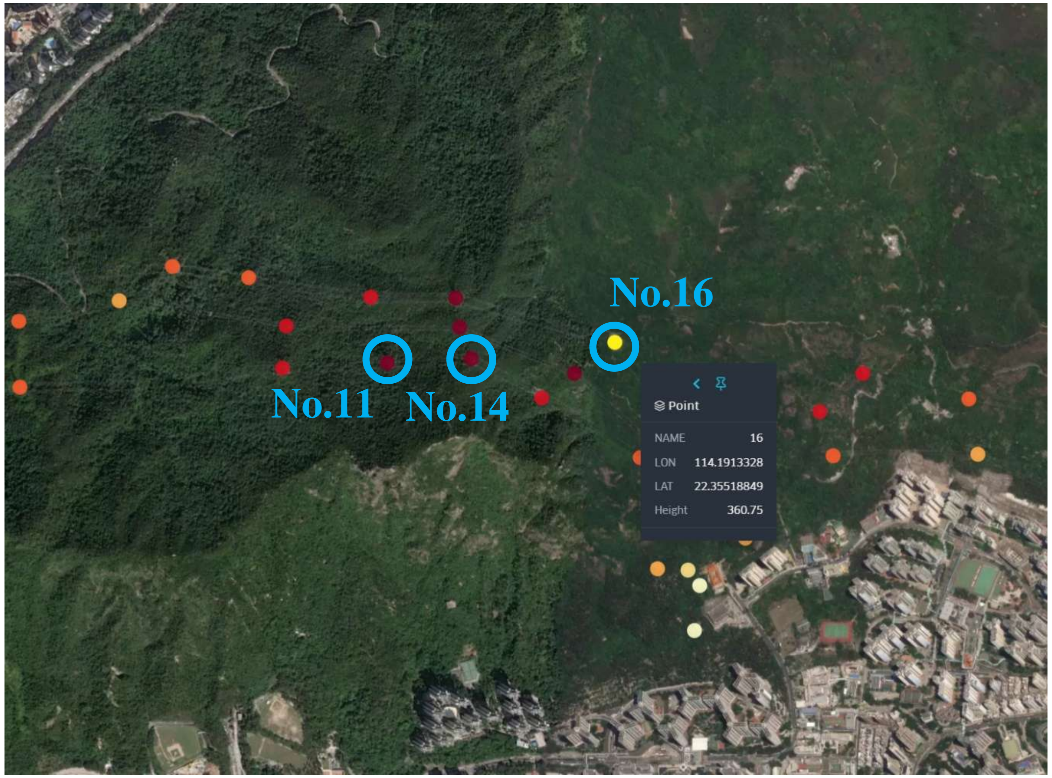


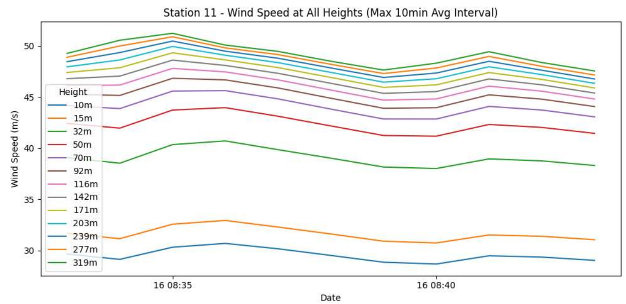


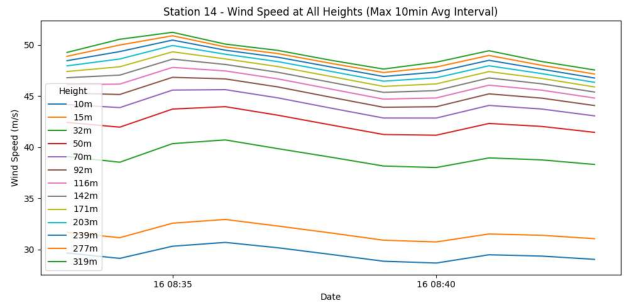


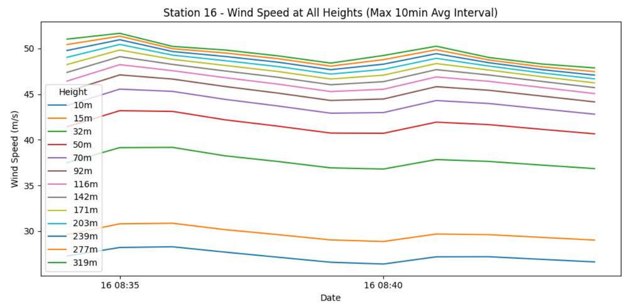
Typhoon-induced reliability Model

WRF Model



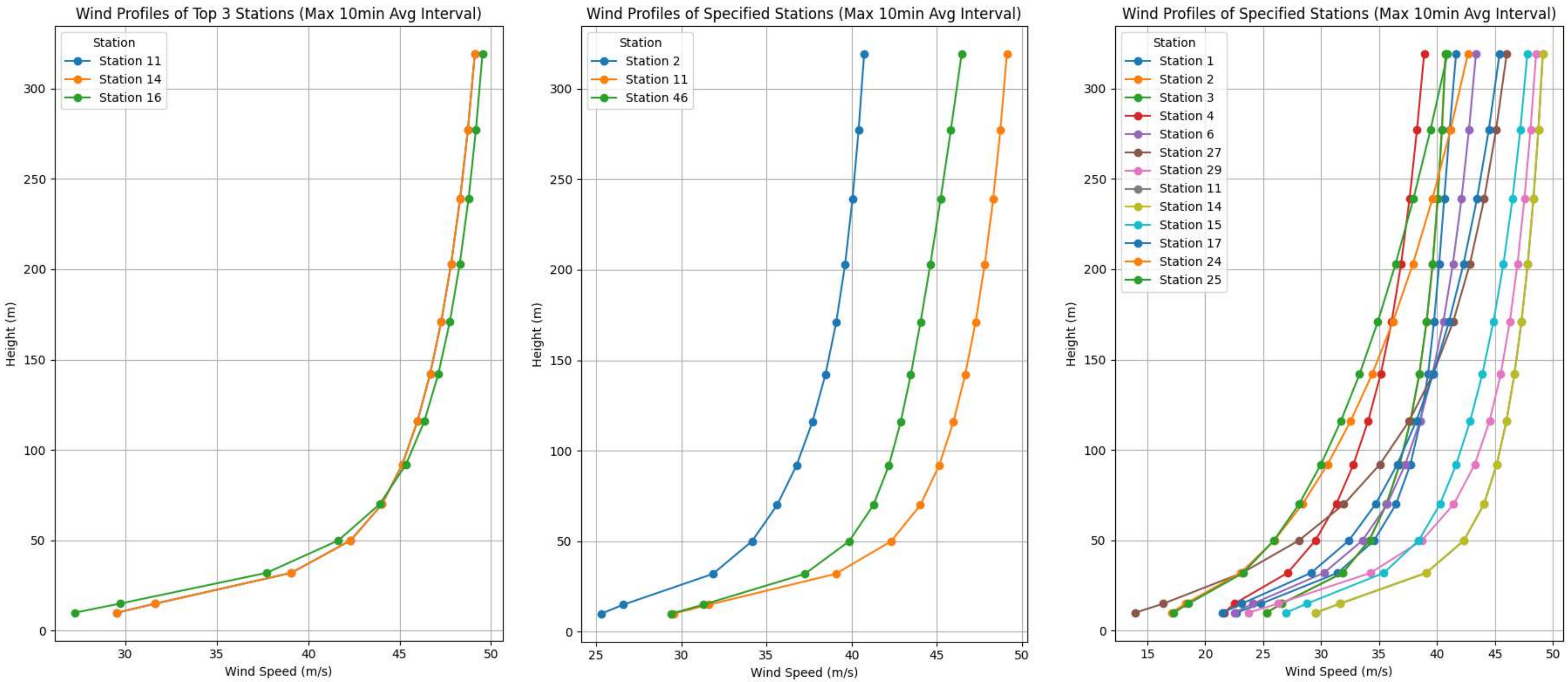






Typhoon-induced reliability Model

WRF Model



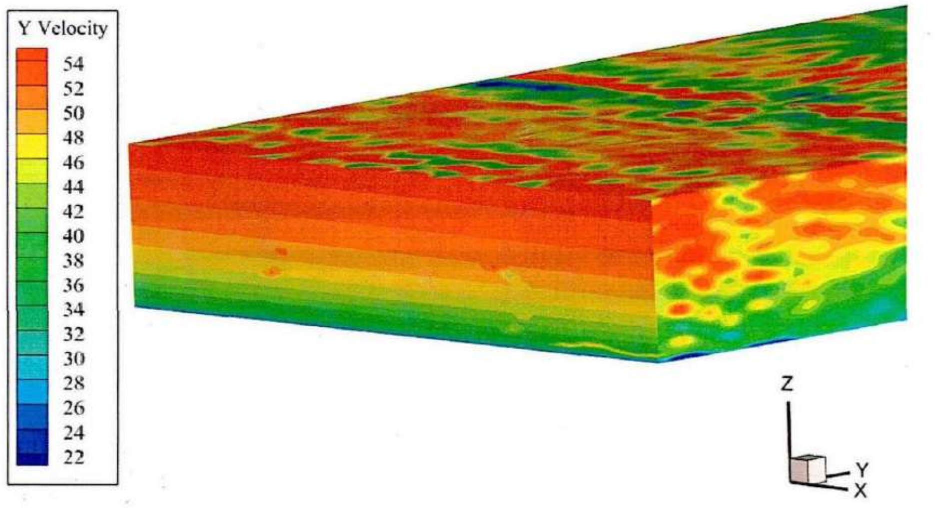
Literature Review

CFD数值模拟方法

台风谱

$$S(f) = 4.8KV_{10}^2\left(\frac{H}{10}\right)^{-0.5\alpha}\frac{x_n}{f(1+x_n^2)^{\frac{7}{8}}}$$

$$x_n = \frac{300fH^\alpha}{V_{10}}$$



Fluent
入口边界条件
出口边界条件
壁面边界条件
对称面边界条件

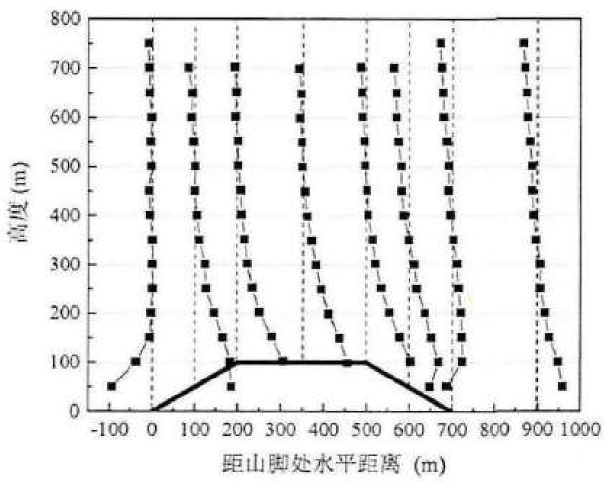
库郎数

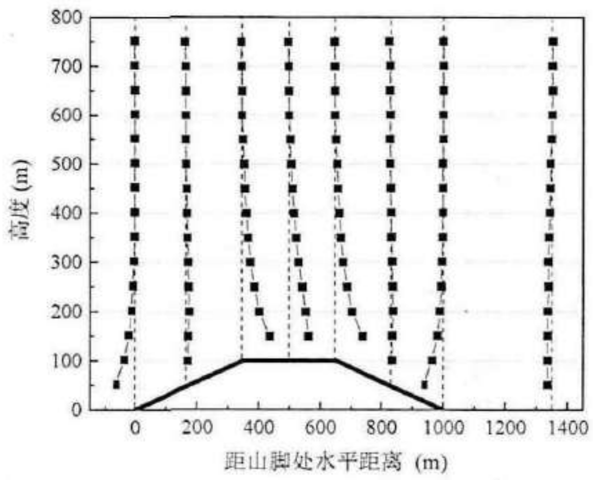
$$CFL = \frac{U\Delta T}{\Delta x} \leq 1$$

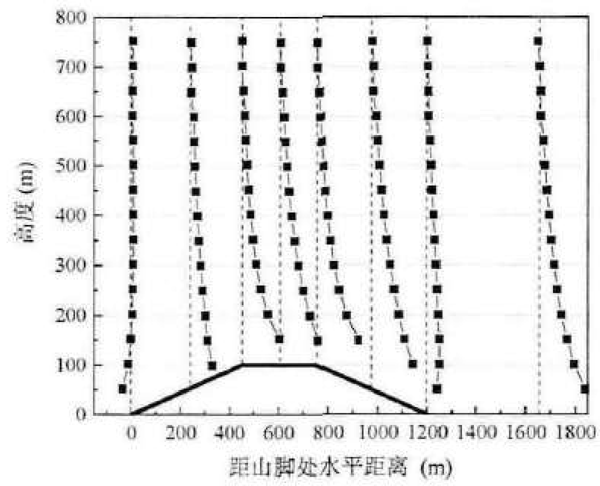
Literature Review

CFD数值模拟方法

不同坡度风加速比



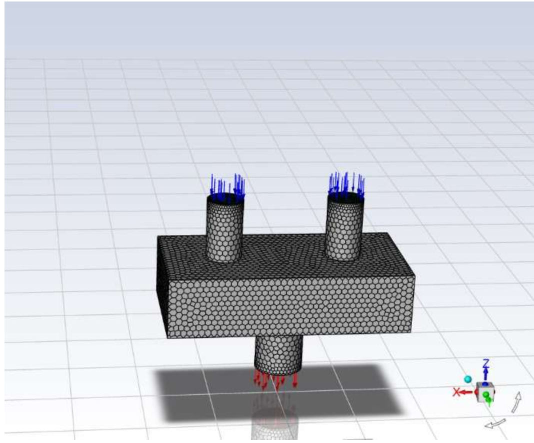


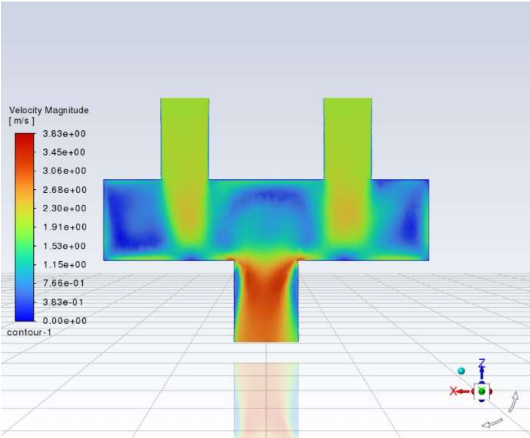


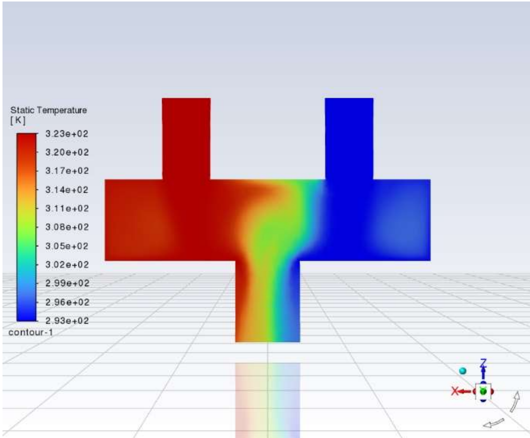
考虑地形影响的台风作用下输电塔－线体系响应分析－山大

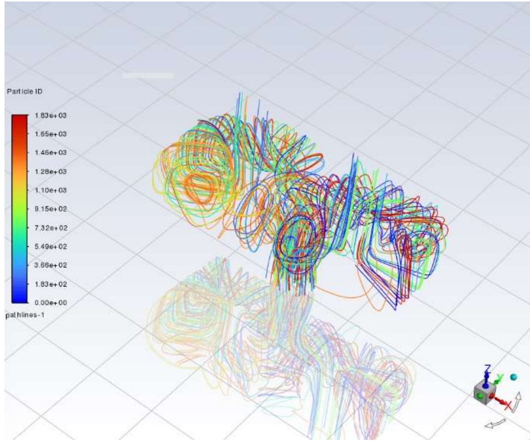
Typhoon-induced reliability Model

CFD Simulation Sample





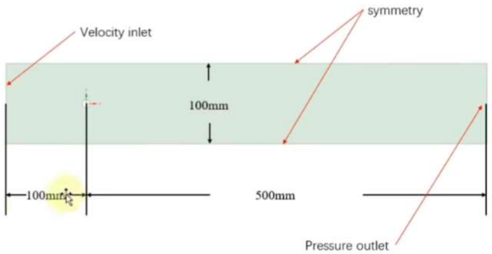


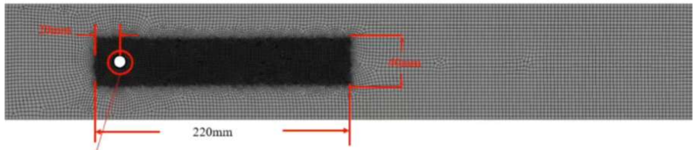


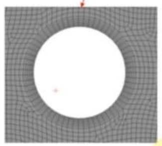
Typhoon-induced reliability Model

CFD Simulation Sample

工况：
雷诺数：100
来流速度：1m/s
密度：10 kg/m³
动力粘度：0.001 Pa*s
圆柱直径：0.01m







网格划分信息
Boi：0.5mm
其他：3mm
圆柱线尺寸：0.5mm
边界层：第一层高度0.1mm，7层，1.15增长率
总数：43724个

$$Sr=\frac{fD}{v}=0.198(1-\frac{19.7}{Re})$$

$$Re=\frac{\rho vD}{\mu}$$

Sr，斯特劳哈尔数；f，涡脱落频率，Hz；D，水力直径，m；v，来流速度，m/s；Re，雷诺数；ρ，密度，kg/m³;μ，动力粘度，Pa*s；

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Thanks

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